

Problem 1

1. $\{K = 0\} = \{\{TTTTTT \dots TT\}\}$ N times.

Since the first tying head appears if there is one head this means that for there to be no typing heads there is no heads.

$A_N = 0$ since there is no heads, so there is no HH pair

$B_N = 0$ since there is no heads, so there is no HT pair

- 2.

Problem 2

Consider the up right lattice paths from $(0,0)$ to $(n, n-k)$.

Let right be if he takes it out of the right pocket and up to be if he takes it out of the left pocket.

Thus, the number of ways to count this is paths from $(0,0)$ to $(n, n-k)$ or $(n-k, n)$ for both pockets thus

$\binom{2n-k}{n} + \binom{2n-k}{n} = 2\binom{2n-k}{n}$ the total number of ways to pick from the left and right pocket is 2^{2n-k} since we have 2 choices for where to take the $2n-k$ matches from

but since we emptied the pocket then checked one more time from the pocket this means we actually check one more time so the total number of possibilities is 2^{2n-k+1}

thus $\mathbb{P}(\text{the first time the mathematician finds a pocket to be empty}) = \frac{2\binom{2n-k}{n}}{2^{2n-k+1}} = \frac{\binom{2n-k}{n}}{2^{2n-k}}$

Problem 3

Let X_i be the indicator function such that $X_i(s) = 1$ if $s(i) = i$ and 0 otherwise

Consider $E[X] = E[X_1 + X_2 + \cdots + X_n] = E[X_1] + \cdots + E[X_n] = \frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n} = 1$

Thus $E[X^2] = E[X_1^2 + X_2^2 + \cdots + X_n^2 + 2X_1X_2 + 2X_1X_3 + \cdots + 2X_{n-1}X_n] = \sum_{i=1}^n E[X_i^2] + 2E[X_1X_2] + 2E[X_1X_3] + \cdots + 2E[X_{n-1}X_n]$

since X_i only outputs 0 or 1 X_i^2 will also output the same thing thus

$= 1 + 2E[X_1X_2] + 2E[X_1X_3] + \cdots + 2E[X_{n-1}X_n] = 1 + 2(E[X_1^2] + E[X_1X_2] + E[X_1X_3] + \cdots + E[X_{n-1}X_n])$

$= 1 + 2\left(\binom{n}{2}E[X_1X_2]\right)$ we can do this since the expected value of two number is the same no matter the two numbers and also there is n choose 2 ways to pick the two pairs of numbers

$E[X_1X_2] = \mathbb{P}(s(1) = 1)\mathbb{P}(s(2) = 2 \mid s(1) = 1)$ there is n choices for the first number and $n - 1$ for the second given 1 is fixed

thus $E[X_1X_2] = \frac{1}{n} \times \frac{1}{n-1}$

$= 1 + 2\left(\binom{n}{2} \frac{1}{n} \times \frac{1}{n-1}\right) = 1 + 2\left(\frac{n(n-1)}{2} \frac{1}{n} \times \frac{1}{n-1}\right) = 1 + 2\left(\frac{1}{2}\right) = 1 + 1 = 2$